

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination (New)

University Theory Examination November-December 2015

PH215 - Pharmacology-I

Date: 01/12/2015, Tuesday

Time: 10.00 a.m. to 1.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section I are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. Which are the targets for drug binding?
[A] Receptor
[B] Enzyme
[C] Ion channel
[D] All of above
2. Which of the following is example of G protein coupled receptor (GPCR)?
[A] Muscarinic receptor
[B] Nicotinic receptor
[C] Cytokine receptor
[D] Oestrogen receptor
3. The characteristic structure of GPCR is made up of
[A] Nine transmembrane alfa helices
[B] Seven transmembrane alfa helices
[C] Three transmembrane beta helices
[D] Five transmembrane beta helices
4. How active protein kinase affecting lipolysis?
[A] Phosphorylase got activated
[B] glycogen synthase got inactivated
[C] glycogen is converted to glucose
[D] lipase got activated
5. Which of the following is not a injectable route of drug administration?
[A] subcutaneous
[B] intravenous
[C] intrathecal
[D] sublingual
6. One of the following drug is an example of Prodrug
[A] Gentamicin
[B] Levodopa
[C] tetracyclin
[D] Morphine

7. What is the toxic metabolite of cyclophosphamide?
 - [A] acrolien
 - [B] fluoride
 - [C] phosphamide
 - [D] tetracyclophosphamide
8. Which food product is avoided with use of tetracycline?
 - [A] Wheat flour
 - [B] Soyabin
 - [C] Milk
 - [D] Eggs
9. What is the effect of combining adrenaline with local anaesthetic injection?
 - [A] absorption of anaesthetic increases
 - [B] vasoconstriction slows absorption of anaesthetics and prolongs local effect
 - [C] vasodilation & reduce local effect
 - [D] no effect of anaesthetic agent observed
10. What are the consequences of drug interaction between warfarin and aspirin?
 - [A] toxicity to liver
 - [B] heart attack
 - [C] risk of bleeding
 - [D] rise in blood pressure
11. Nicotinic Ach receptors are present in following organ system except
 - [A] liver
 - [B] Muscle
 - [C] Ganglion
 - [D] CNS
12. One of the following drug is an example of muscarinic agonist
 - [A] Atropine
 - [B] Oxybutynin
 - [C] Carbachol
 - [D] pirenzapine
13. M2 receptor producing major effect on following system
 - [A] Cardiac system
 - [B] Central nervous system
 - [C] Liver
 - [D] Glandular system
14. Following which third generation beta blocker having anti proliferative and antioxidant property apart from NO releasing vasodilation property?
 - [A] Carvedilol
 - [B] Nebivolol
 - [C] Labetolol
 - [D] Celiprolol
15. Dobutamine is used in cardiogenic shock due to
 - [A] Alpha 1 action
 - [B] Beta 1 action
 - [C] Beta 2 action
 - [D] Alpha 2 action
16. Which of the following sympathomimetics is a beta2-selective agonist?
 - [A] Isoproterenol
 - [B] Xylometazoline
 - [C] Dobutamine
 - [D] Terbutalin

17. What are the major effects of atropine on Eyes?
 [A] Myosis
 [B] Drieness of Eye
 [C] Mydriasis & Cycloplegia
 [D] All of above
18. One of the following is not a clinical use of *alpha adrenergic antagonist*.
 [A] Cardiac arrest
 [B] Severe Hypertension
 [C] Benign Prostatic hypertrophy
 [D] Phaeochromocytoma
19. What are the major clinical uses of Beta adrenergic antagonist.
 [A] Angina pectoris & glaucoma
 [B] Myocardial infarction & heart failure
 [C] Both A & B
 [D] Liver disease
20. Salbutamol is of which class drug and useful in which condition.
 [A] α_1 agonist & nasal congestion
 [B] β_2 agonist and Asthma
 [C] α_2 agonist & glaucoma
 [D] β_1 antagonist and Heart failure

SECTION – II

Q 2 Attempt any FOUR of the following

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|---|---|----|
| A | Explain concept of Cholinergic and Noradrenergic transmission (draw chart) and provide details of drugs acting on each pathways | 05 |
| B | Explain Characteristics of Neuromuscular blocking drugs | 05 |
| C | Discuss in short about drugs affecting Muscarinic receptors | 05 |
| D | Discuss in short about drugs affecting adrenergic receptors | 05 |
| E | Provide clinical uses of Alpha agonists, Alpha antagonist and Beta antagonist | 05 |
| F | Provide detail note on Noradrenaline synthesis and release. | 05 |

SECTION – III

Q 3 Attempt any FOUR of the following

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|---|---|----|
| A | Explain the following terms
1. Iatrogenic disease
2. Pharmacokinetic
3. Bioavailability
4. Biotransformation
5. Volume of distribution | 05 |
| B | What is Receptor? Explain signal transduction mechanism of G Protein coupled receptor | 05 |
| C | Write a short note on Clinical trials. | 05 |
| D | Explain the concept of Bioavailability and drug excretion in short. | 05 |

- E Discuss techniques for administration of Local anesthetic agents 05
- F Write a note on the following 05
1. Write a note on Anticholinesterase drugs
 2. Ganglionic stimulants
- Q 4 Attempt any FOUR of the following**
- A Classify Routes of Drug administration. Explain advantages and disadvantages of Parenteral route of administration. 05
- B Write a detailed note on Plasma protein binding of a drug. 05
- C Provide a brief note on drug antagonism. Enlist factors affecting Gastrointestinal absorption. 05
- D Explain concept of Drug metabolism in detail. 05
- E Provide a note on different targets for drug action and explain any one in short. 05
- F Explain concept of drugs interaction in detail. 05